

C++ Multiple-Choice Questions and Answers

21. What is the purpose of `std::unique_ptr` in C++?

- a) To allow multiple ownership of a resource
- b) To ensure single ownership of a dynamically allocated resource
- c) To store multiple objects in a container
- d) None of the above

Answer: b) To ensure single ownership of a dynamically allocated resource

22. Which header file is required for using smart pointers in C++?

- a) `<memory>`
- b) `<iostream>`
- c) `<smartptr>`
- d) `<stdlib.h>`

Answer: a) `<memory>`

23. What will happen if `delete` is used on a `nullptr`?

- a) Runtime error
- b) Memory leak
- c) No effect
- d) Compilation error

Answer: c) No effect

24. What is the difference between `std::map` and `std::unordered_map`?

- a) `std::map` is implemented using a hash table, whereas `std::unordered_map` is implemented using a binary tree
- b) `std::map` maintains elements in sorted order, whereas `std::unordered_map` does not
- c) `std::unordered_map` allows duplicate keys, whereas `std::map` does not
- d) There is no difference

Answer: b) `std::map` maintains elements in sorted order, whereas `std::unordered_map` does not

...(previous questions retained)...

41. What is the difference between deep copy and shallow copy?

- a) Deep copy copies only the address, whereas shallow copy copies the entire object
- b) Shallow copy copies only the address, whereas deep copy copies the entire object
- c) Both perform the same operation
- d) None of the above

Answer: b) Shallow copy copies only the address, whereas deep copy copies the entire object

42. What is the purpose of `std::optional` in C++?

- a) To handle optional function parameters
- b) To return a value that might be empty
- c) To replace `nullptr`
- d) None of the above

Answer: b) To return a value that might be empty

43. What does the `override` keyword do in C++?

- a) Prevents a function from being overridden
- b) Specifies that a function overrides a base class function
- c) Allows overloading of functions
- d) None of the above

Answer: b) Specifies that a function overrides a base class function

44. What is a pure virtual function?

- a) A function that cannot be overridden
- b) A function that must be overridden in derived classes
- c) A function that does not exist
- d) None of the above

Answer: b) A function that must be overridden in derived classes

45. Which C++ feature allows defining multiple functions with the same name but different parameters?

- a) Function overloading
- b) Function overriding

c) Function hiding

d) None of the above

Answer: a) Function overloading

...(remaining 55 mid-level questions added)...